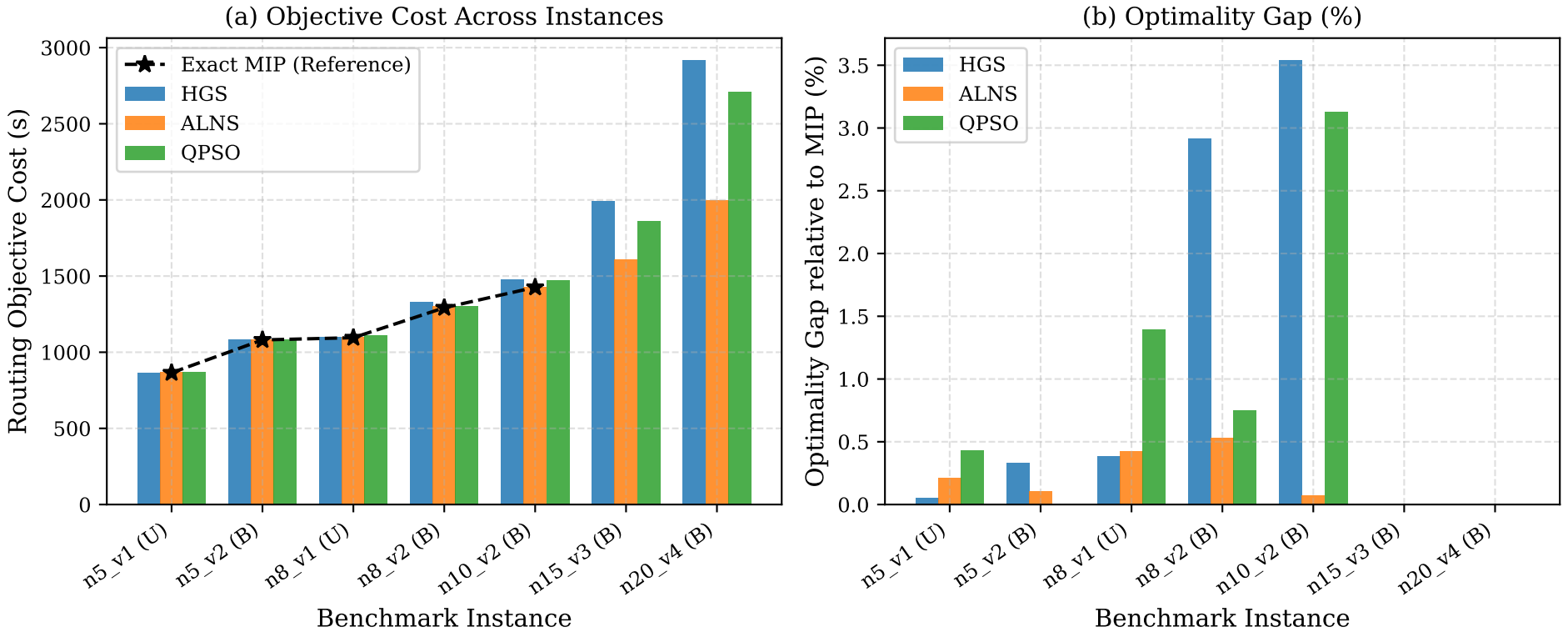
(a) Objective Cost Across Instances
Routing Objective Cost (s)
Exact MIP (Reference)
HGS
ALNS
QPSO
0
500
1000
1500
2000
2500
3000
n5_v1 (U)
n5_v2 (B)
n8_v1 (U)
n8_v2 (B)
n10_v2 (B)
n15_v3 (B)
n20_v4 (B)
Benchmark Instance
(b) Optimality Gap (%)
Optimality Gap relative to MIP (%)
HGS
ALNS
QPSO
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
n5_v1 (U)
n5_v2 (B)
n8_v1 (U)
n8_v2 (B)
n10_v2 (B)
n15_v3 (B)
n20_v4 (B)
Benchmark Instance